Not All GPU Hours Are Created Equal

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Silicon Data's indices rest on a rated GPU hour being a known, interchangeable quantity. I spent two days renting and measuring GPUs to test that, and it breaks on one variable: the host the card runs in. Closing that gap lets a rating carry more weight when settling a futures contract, deciding an RMA claim, or backing a loan.

What the research showed

Three things decide how much work a GPU actually delivers, and neither its model name nor its grade specifies them.

1. **"H100" ships in different power envelopes, and you can't see which before renting.**

   On Runpod, six H100 SXM cards ran at 700 W, while both H100 NVL cards I rented arrived at 310 W, below NVIDIA's published 350 to 400 W range. Runpod, Vast.ai, and Lambda show no power limit before rental. At that setting, the NVL costs 45% more per output token than the SXM, despite having more memory bandwidth.

2. **The host's temperature determines how accurately a short test predicts sustained performance.**

   Ratings come from short tests, because benchmarking time is time a GPU isn't earning. On three L40S cards, a 60-second test read about 8.6% high on two hosts that settled near 79 °C, and 2.7% high on one at 68 °C. H100s stayed under 1%.

3. **The host's temperature stability determines whether a rating repeats.**

   An L40S in a steady host returned the same result four times, within 0.06%. One in a host that drifted 7 °C spread 3.85%, more than double the gap between two different H100s. On an unstable host, a retest can't tell a worse card from a warmer hour.

Per clock, cards of the same chip matched within 0.1%, so every difference above comes from the host.

What a rating should record

QuickMark already records power and temperature, so most of this is a display change. The grade stays; these fields tell a buyer how far to trust it.

Field	Why it matters	Who it protects
Power limit and its default	Two "H100s" can differ 2.26× in power and 45% in cost per token	Futures contracts that treat one H100 hour as interchangeable
Loaded temperature	Sets how far a short test overstates	Teams validating clusters before turn-up; procurement teams in RMA disputes
Repeat spread	Shows whether the number would come out the same twice	Lenders marking collateral; resellers grading inventory

Suggested Changes

1. **Show the number and conditions behind each grade.**

   Publish the measured value and its range beside the letter, plus the test temperature and power limit. A buyer can then tell apart two cards with the same grade, and when a retest comes in lower, see whether the card or the host changed. QuickMark already records power and temperature, so this is mostly a display change.

2. **Price delivered work, not just the card name.**

   Joining SiliconMark results to rental records would let buyers compare offers per unit of work delivered, and give contracts a clear standard for what an H100 hour must provide.

3. **Start a drift study on reserved devices.**

   The open question is how long a certificate holds once a card changes hands. At Catalog, I designed an accelerated decay study to answer the same question for DNA storage and computation.

Experiment Details

I benchmarked three L40S cards and three H100 variants on Runpod, sampling clocks, power, and temperature every 500 ms, and checked power limits on nine H100s. Predictions were committed to git before the runs that tested them. QuickMark graded one L40S D whether started cold, warm, or recooled, so it handles a cold start well. Open question: grades are fleet quartiles, and my three L40S cards spanned 8%, so if a quartile is narrower than about 3%, host conditions alone can move a card a full letter.

Limits: one provider, a few cards per model, hours of testing. Code and raw data: github.com/maker-sean/gpubench.